

- Use the radial wave function for the 3p orbital of a hydrogen atom (see Table 5.2) to calculate the value of r for which a node exists.
 - Find the values of r for which nodes exist for the 3s wave function of the hydrogen atom.
- The helium ion He^+ is a one-electron system whose wave functions and energy levels are obtained from those for H by changing the atomic number to $Z = 2$. Calculate the average distance of the electron from the nucleus in the 2s orbital and in the 2p orbital.
- Write ground-state electron configurations for the ions Li^- , B^+ , F^- , Al^{3+} , S^- , Ar^+ , Br^+ , and Te^- . Which do you expect to be paramagnetic due to the presence of unpaired electrons?
- Photoelectron spectroscopy studies of silicon atoms excited by X-rays with wavelength $9.890 \times 10^{-10} \text{ m}$ show four peaks in which the electrons have speeds $2.097 \times 10^7 \text{ m s}^{-1}$, $2.093 \times 10^7 \text{ m s}^{-1}$, $2.014 \times 10^7 \text{ m s}^{-1}$, and $1.971 \times 10^7 \text{ m s}^{-1}$. (Recall that $1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2}$.)
 - Calculate the ionization energy of the electrons in each peak.
 - Assign each peak to an orbital of the silicon atom.
- Photoelectron spectroscopy studies have determined the orbital energies for chlorine atoms to be

1s -2,835 eV

2s -273 eV

2p -205 eV

3s -21 eV

3p -10 eV

Estimate the value of Z_{eff} for Cl in each of these orbitals.

TABLE 5.2
Angular and Radial Parts of Wave Functions for One-Electron Atoms

Angular Part $Y(\theta, \phi)$	Radial Part $R_{n\ell}(r)$
$\ell = 0 \left\{ Y_s = \left(\frac{1}{4\pi} \right)^{1/2} \right.$	$R_{1s} = 2 \left(\frac{Z}{a_0} \right)^{3/2} \exp(-\sigma)$ $R_{2s} = \frac{1}{2\sqrt{2}} \left(\frac{Z}{a_0} \right)^{3/2} (2 - \sigma) \exp(-\sigma/2)$ $R_{3s} = \frac{2}{81\sqrt{3}} \left(\frac{Z}{a_0} \right)^{3/2} (27 - 18\sigma + 2\sigma^2) \exp(-\sigma/3)$
$\ell = 1 \left\{ \begin{array}{l} Y_{p_x} = \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \cos \phi \\ Y_{p_y} = \left(\frac{3}{4\pi} \right)^{1/2} \sin \theta \sin \phi \\ Y_{p_z} = \left(\frac{3}{4\pi} \right)^{1/2} \cos \theta \end{array} \right.$	$R_{2p} = \frac{1}{2\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma \exp(-\sigma/2)$ $R_{3p} = \frac{4}{81\sqrt{6}} \left(\frac{Z}{a_0} \right)^{3/2} (6\sigma - \sigma^2) \exp(-\sigma/3)$
$\ell = 2 \left\{ \begin{array}{l} Y_{d_{x^2}} = \left(\frac{5}{16\pi} \right)^{1/2} (3 \cos^2 \theta - 1) \\ Y_{d_{xz}} = \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \cos \phi \\ Y_{d_{yz}} = \left(\frac{15}{4\pi} \right)^{1/2} \sin \theta \cos \theta \sin \phi \\ Y_{d_{xy}} = \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \sin 2\phi \\ Y_{d_{x^2-y^2}} = \left(\frac{15}{16\pi} \right)^{1/2} \sin^2 \theta \cos 2\phi \end{array} \right.$	$R_{3d} = \frac{4}{81\sqrt{30}} \left(\frac{Z}{a_0} \right)^{3/2} \sigma^2 \exp(-\sigma/3)$
$\sigma = \frac{Zr}{a_0} \quad a_0 = \frac{\epsilon_0 h^2}{\pi e^2 m_e} = 0.529 \times 10^{-10} \text{ m}$	